

Missing Values

Charles River: Shipment table (Order-to-Cash)

ShipmentNumber	ShipmentDate	FreightCost	TrackingNumber
SH-2024-000028	2024-01-15	\$189.01	TRK202400000028
SH-2024-000029	2024-01-16	\$35.31	(blank)
SH-2024-000049	2024-01-21	\$106.73	(blank)

Completeness problem: TrackingNumber is blank, so there is no carrier record for these shipments.

Inconsistent Values

Charles River: Shipment table (Order-to-Cash)

ShipmentNumber	ShipmentDate	Status	DeliveryDate
SH-2024-000001	2024-01-03	Delivered	2024-01-08
SH-2024-000004	2024-01-06	In Transit	2024-01-10
SH-2024-000038	2024-01-18	In Transit	2024-01-20

Consistency problem: These shipments are still marked In Transit, although their DeliveryDate passed years before the data was extracted.

Duplicate Records

Charles River: PurchaseInvoice table (Procure-to-Pay)

SupplierID	InvoiceNumber	InvoiceDate	GrandTotal
2	V0002-2024-000182	2024-02-29	\$4,152.60
2	V0002-2024-000182	2024-03-04	\$3,454.20
2	V0002-2024-000204	2024-03-17	\$950.16

Duplicate: The same supplier invoice number is recorded twice, with different dates and amounts. This is a duplicate-payment risk that requires investigation.

Outliers

Charles River: Shipment table (Order-to-Cash)

ShipmentNumber	WarehouseID	FreightCost
SH-2024-000001	2	\$98.03
SH-2024-000002	1	\$152.57
SH-2024-000003	2	\$248.92
SH-2026-008319	2	\$873.95

Outlier: A FreightCost of \$873.95 is several times the average shipment (about \$150). It may be a legitimate large or rush shipment, so investigate before treating it as an error.